12.21 Breaking a Stick

WORKSHEET

The question

STEP III 1990, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s3-q16

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Three lengths form a triangle if and only if each is less than the sum of the other two. If the lengths add up to L, what does this say about the longest one? When is the angle opposite side c obtuse?

STAGE 1 One cut, then halve

Covers: Part (i)

Ask yourself Three lengths that add to 1 form a triangle exactly when each is less than $\frac{1}{2}$. What does that say about X?

Pause and try The triangle is isosceles. When is the angle opposite X obtuse? Use the cosine rule. Given that a triangle forms, X is uniform on $[0, \frac{1}{2}]$.

STAGE 2 The second construction

Covers: Part (ii), the construction and the condition

Ask yourself Check the three triangle inequalities for sides x, y and $2 - x - y$ on the given region.

Ask yourself The angle at B is opposite the side of length y. Write the condition for it to be obtuse as an inequality for y.

STAGE 3 The area

Covers: Part (ii), the probability

Pause and try Divide $x^2 - 2x + 2$ by $2 - x$. Find the area of the part of the region above the curve, and divide by the area of the whole region.

What's really going on

The classic broken stick

Break a stick at two independent uniform points U and V. The pair (U, V) is uniform on the unit square, and the three pieces form a triangle exactly when each is less than $\frac{1}{2}$: that cuts out four small triangles of total area $\frac{3}{4}$, leaving probability $\frac{1}{4}$. Given that a triangle forms, the probability that it is acute is $3\ln 2 - 2 \approx 0.079$ out of $\frac{1}{4}$, that is about 32%. The pieces are an example of a *uniform random partition*, and their joint distribution is the *Dirichlet distribution*, which appears in Bayesian statistics and in models of how a resource is shared out at random.

How you break the stick matters

Notice that the two methods in the question give different answers, $3 - 2\sqrt{2}$ and $3 - 4\ln 2$, to similar-sounding questions, and both differ from the classic method. As with Bertrand's chords (Section 12.3), "random" means nothing until the procedure is specified.

Random triangles

Problems about random triangles were popular Victorian puzzles. Lewis Carroll's *Pillow Problems* (1893) asks for the probability that a triangle chosen "at random" is obtuse, and his answer, like those in this section, depends entirely on the model of randomness he chose. Problems about random triangles have continued to produce surprises: for instance, the probability that three uniformly random points in a square form an obtuse triangle is about 0.725, computed exactly only in the twentieth century.

Try this next

1. **The classic.** Show that if a stick is broken at two independent uniform points, the probability that the pieces form a triangle is $\frac{1}{4}$.

2. **Break the longer piece.** Break a stick at a uniform point, then break the *longer* piece at a uniform point. Show that the probability that the three pieces form a triangle is $2\ln 2 - 1 \approx 0.386$.

3. **Acute or obtuse.** In part (ii), find the probability that the triangle ABC has an obtuse angle at C (opposite $AB = 2 - x - y$). Hence find the probability that it has an obtuse angle somewhere.

4. **Expected longest piece.** For the classic broken stick, show that the expected length of the longest piece is $\frac{11}{18}$.

Further reading

- **Martin Gardner, *The Colossal Book of Mathematics*** (Norton), which includes the broken-stick problem.
- **Frederick Mosteller, *Fifty Challenging Problems in Probability*** (Dover).
- **Lewis Carroll, *Pillow Problems*** (Dover, with *A Tangled Tale*). Victorian puzzles, including random triangles.

The full guided solution is in Section 12.21:
https://beyondstep.keithmcnulty.org/ch12/broken-stick.html